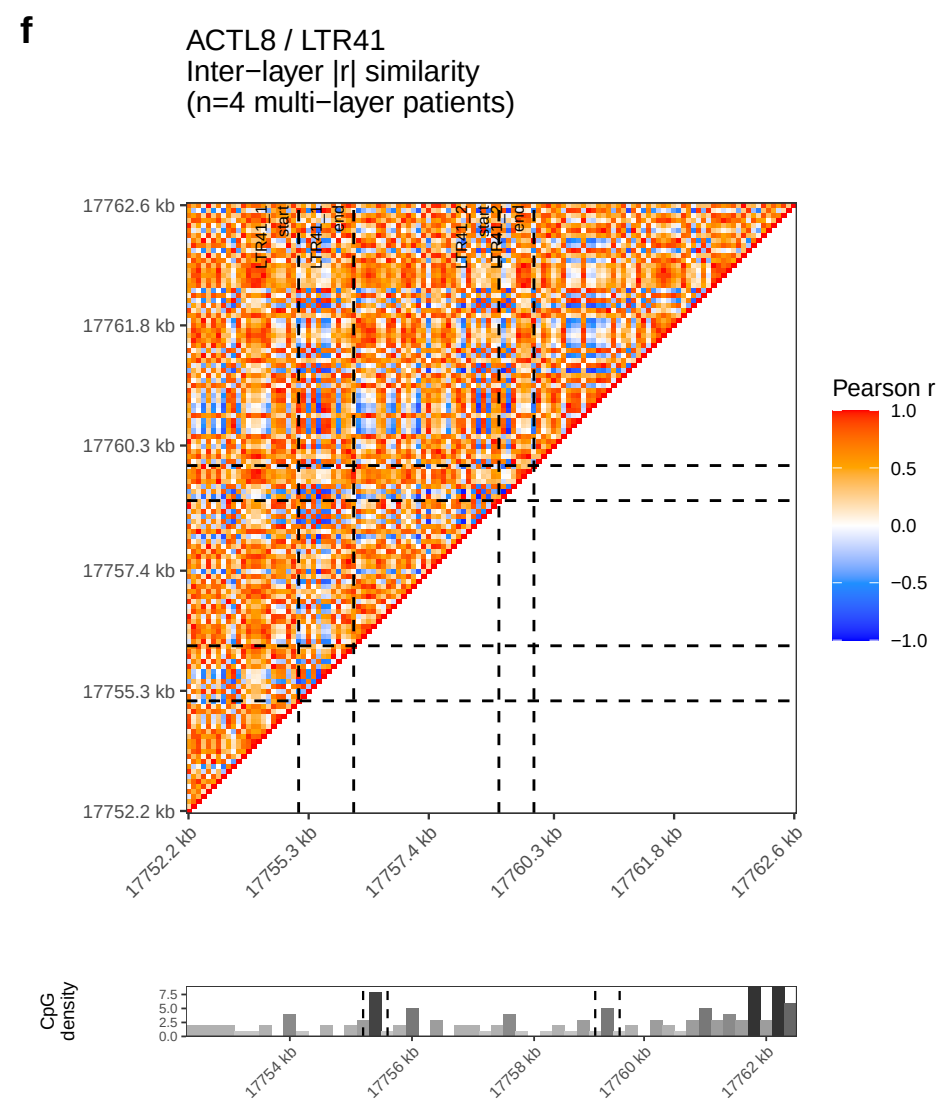
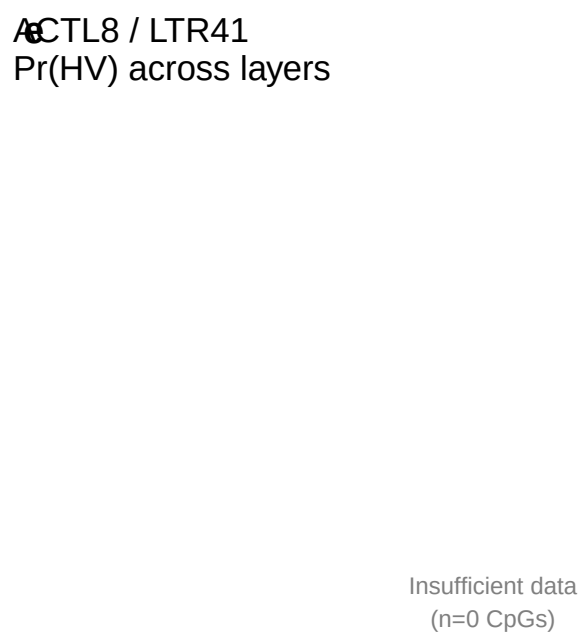
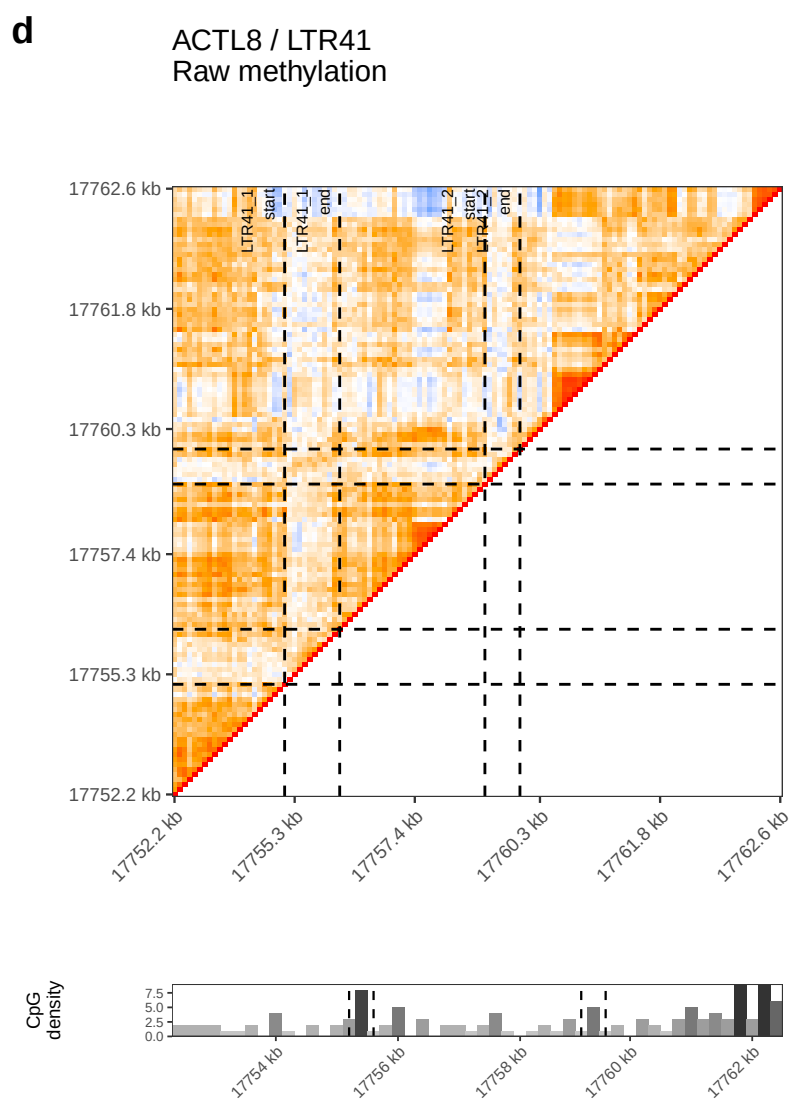
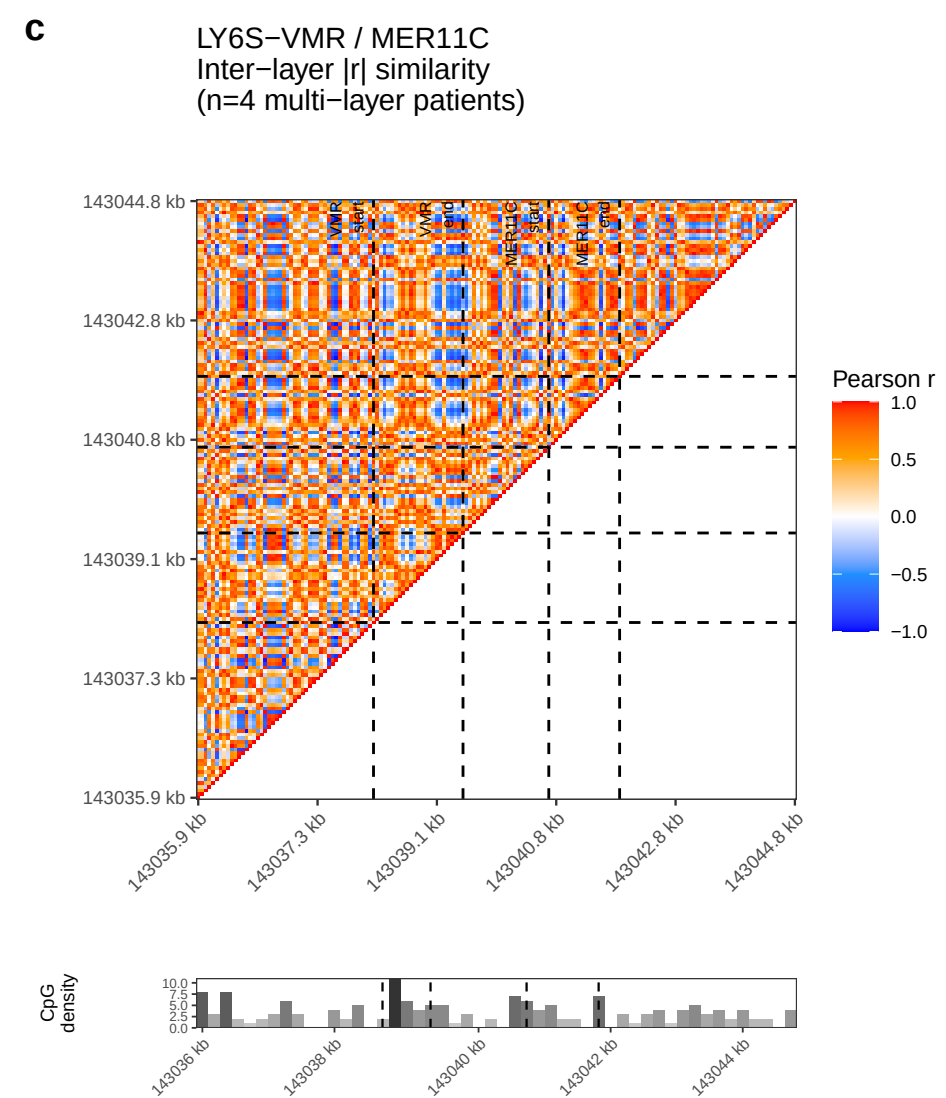
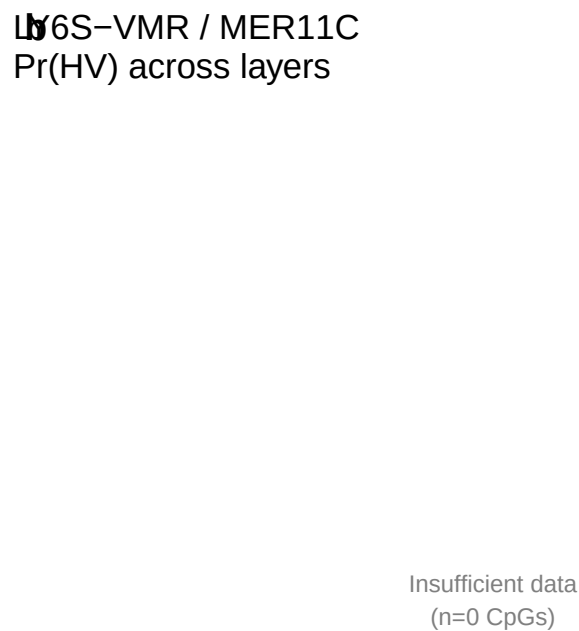
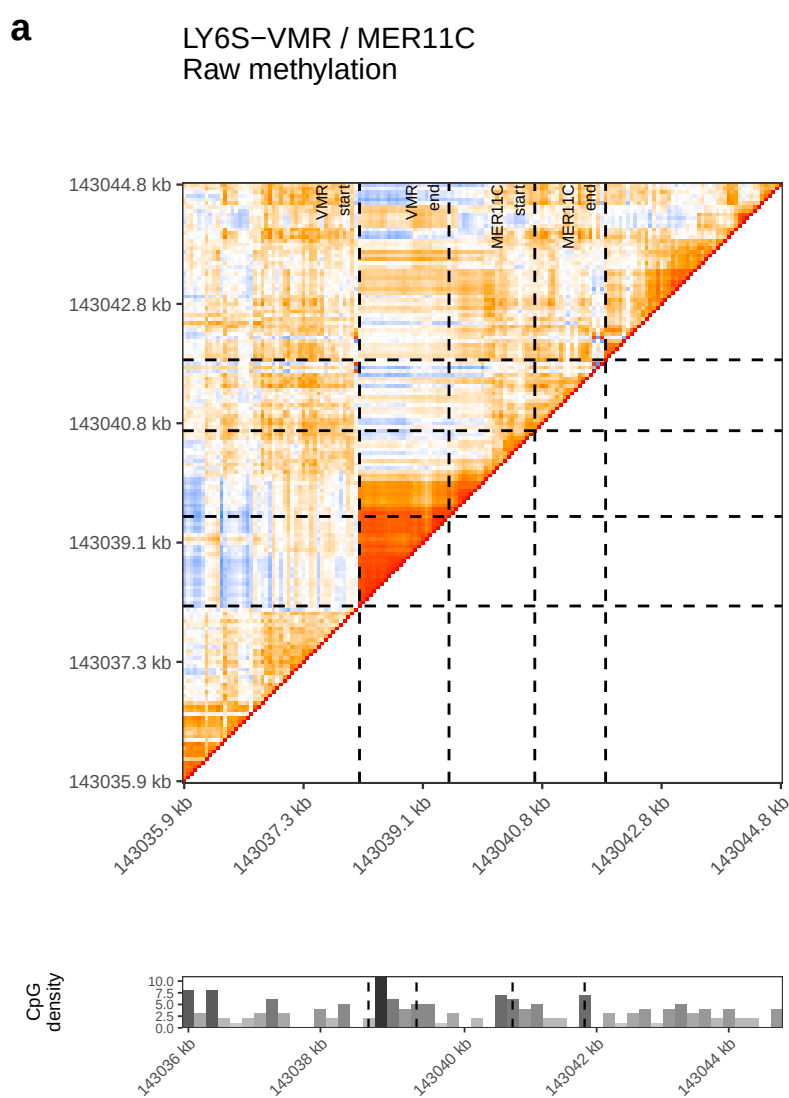




Pairwise Pearson  $r$  between CpG sites (upper triangle, rank-ordered by position).

a/d: Raw methylation –  $r$  computed across 135 patients (mean per patient per CpG). High  $r$  = CpGs co-methylated across individuals (epiallele block). Negative  $r$  = opposing methylation states.

b/e: Pr(HV) –  $r$  across 3 germ layers; high  $r$  = CpGs co-hypervariable in same layers (ME signature).

c/f: Inter-layer  $|r|$  similarity – pairwise similarity of per-CpG Endo-Meso  $r$  values across 4 multi-layer patients; high  $r$  = CpGs with concordant inter-layer methylation behaviour.

Colour: blue = negative  $r$ ; white = 0; orange/red = positive  $r$ . Black dashed = feature boundaries. Bottom strip = CpG density across the window.